Bull Math Biol (2012) 74:1727–1753
DOI 10.1007/s11538-012-9733-1

ORIGINAL ARTICLE

Modeling of Hysteresis in Gene Regulatory Networks

J. Hu · K.R. Qin · C. Xiang · T.H. Lee

Received: 15 July 2011 / Accepted: 30 April 2012 / Published online: 17 May 2012

Abstract Hysteresis, observed in many gene regulatory networks, has a pivotal impact on biological systems, which enhances the robustness of cell functions. In this paper, a general model is proposed to describe the hysteretic gene regulatory network by combining the hysteresis component and the transient dynamics. The Bouc–Wen hysteresis model is modified to describe the hysteresis component in the mammalian gene regulatory networks. Rigorous mathematical analysis on the dynamical properties of the model is presented to ensure the bounded-input-bounded-output (BIBO) stability and demonstrates that the original Bouc–Wen model can only generate a clockwise hysteresis loop while the modified model can describe both clockwise and counter clockwise hysteresis loops. Simulation studies have shown that the hysteresis loops from our model are consistent with the experimental observations in three mammalian gene regulatory networks and two *E.coli* gene regulatory networks, which demonstrate the ability and accuracy of the mathematical model to emulate natural gene expression behavior with hysteresis. A comparison study has also been conducted to show that this model fits the experiment data significantly better than previous ones in the literature. The successful modeling of the hysteresis in all the five hysteretic gene regulatory networks suggests that the new model has the potential to be a unified framework for modeling hysteresis in gene regulatory networks

J. Hu · C. Xiang (✉) · T.H. Lee
Department of Electrical and Computer Engineering, National University of Singapore, Singapore 117576, Singapore
e-mail: elexc@nus.edu.sg

J. Hu · T.H. Lee
Graduate School for Integrative Science and Engineering, National University of Singapore, Singapore, Singapore

K.R. Qin
Department of Biomedical Engineering, Dalian University of Technology, Dalian 116024, China

and provide better understanding of the general mechanism that drives the hysteretic function.

Keywords Gene regulatory network · Hysteresis · Modified Bouc–Wen model · Transient response · Positive Feedback · BIBO stability

1 Introduction

Hysteresis is a ubiquitous phenomenon that occurs in diverse disciplines ranging from electronics to economics, from mechanics to life science (Tan and Iyer 2009). Roughly speaking, hysteresis refers to the phenomenon that the response of a system takes on different values for an increasing input than for a decreasing one. In the past decade, a number of experimental studies have observed the hysteretic behavior in a wide range of subjects in natural biology such as gene regulatory networks and cell cycles (Angeli et al. 2004; Atkinson et al. 2003; Becskei et al. 2001; Becskei and Serrano 2000; Justman et al. 2009; Ferrell et al. 2009; Bagowski and Ferrell 2001; Ozbudak et al. 2004; Xiong and Ferrell 2003). It has been demonstrated that hysteresis has vital impact on biological systems as it provides a mechanism that enhances the robustness of cell functions against random perturbations (Solomon 2003; Sha et al. 2003; Pomerening et al. 2003; Han et al. 2005).

With the rapid accumulation of genetic information as well as development of biotechnology, studies on the synthetic gene network move from bacteria or lower eukaryotes (Chang et al. 2010; Maeda and Sano 2006) to mammalian systems (Kramer et al. 2004; Kramer and Fussenegger 2005; Greber and Fussenegger 2007; Weber et al. 2007; May et al. 2008; Longo et al. 2010). Hysteresis is observed in these synthetic gene regulatory networks. The threshold of the inducer concentration for gene expression level in the gene networks to switch from one state to another takes on different values for an increasing inducer concentration than that for a decreasing one. For instance, the existence of hysteresis is verified in a mammalian gene regulatory network (Kramer and Fussenegger 2005). The molecular configuration of the mammalian gene regulatory network with hysteresis is shown in Fig. 1 (Kramer and Fussenegger 2005), capitalizing the interconnection of tetracycline- and macrolide-responsive transgene control modalities. The clinically licensed antibiotic erythromycin (EM) modulates the activity of the transactivator (TA) through bolishing the transrepressor capacity, which also binds to the hybrid promoter with an inhibitory effect. The human placental secreted alkaline phosphatase (SEAP) is expressed cocistronically with TA. Therefore, varying $[EM]$ (the square bracket [.] denotes concentration) results in different active transrepressor concentrations and can be used to regulate $[SEAP]$. The experiment data are shown in Fig. 2, which indicates that when $[EM]$ was little in the historical cultivation medium, it requires 1000 ng/mL $[EM]$ to switch SEAP production from an OFF state to an ON state, whereas it only requires 500 ng/mL $[EM]$ to switch $[SEAP]$ from an ON state to an OFF state when the historical cultivation medium contained high $[EM]$. The significant advantage of the gene regulatory network with hysteresis is its inherent ability to buffer against modest changes in the inducer concentration. Hence, minor fluctuations in inducer

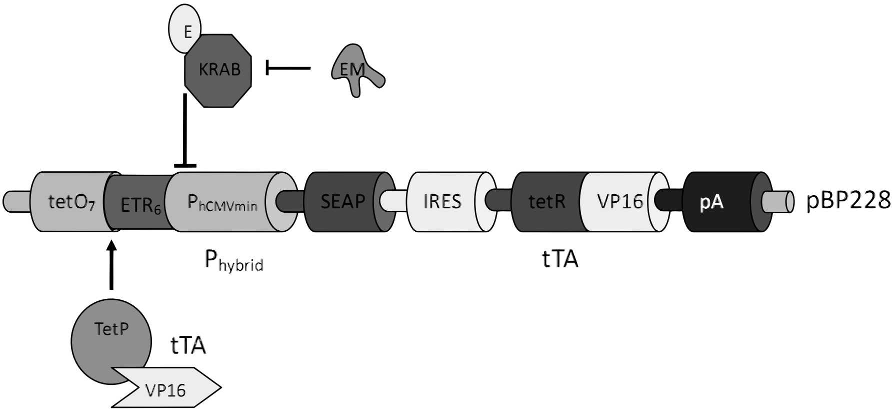
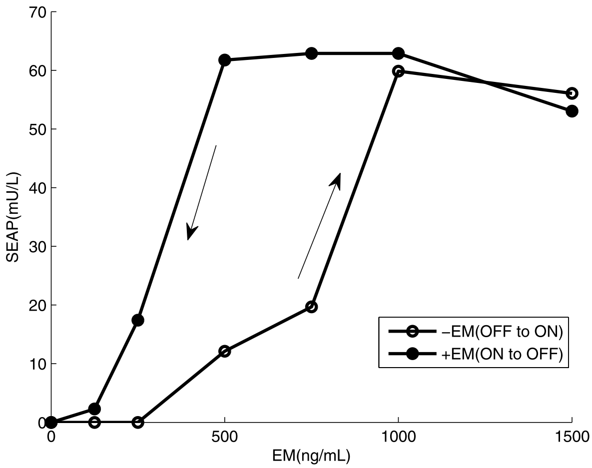


Fig. 1 Molecular configuration of the synthetic hysteretic mammalian gene regulatory network

Fig. 2 Hysteresis loop observed in mammalian gene regulatory network



concentration close to the threshold value to turn on the gene expression will keep the gene expression at ON state as the inducer concentration is far away from the threshold to turn off the gene expression (Greber and Fussenegger 2007). The above mentioned experimental data for hysteresis were measured only after enough time for gene expression to reach the steady state. In recent study, transient response before the gene expression reaching steady state value was also investigated by single-cell measurement (Longo et al. 2010). It is evident from the experiment observation that above the bistable regime where all the gene expression will eventually reach the ON state, the stronger positive feedback by increasing the inducer concentration results in a shorter delay time prior to steady state at ON state. This experimentally observed dynamical behavior of coherent activation is important for drawing insights into biological processes that rely on positive feedback regulation.

Biological systems are highly complex while mathematical models can represent a quantitative system-level understanding of the cellular process. A mathematical model was proposed by Kramer and Fussenegger (KF model) to depict the synthetic hysteretic mammalian gene regulatory network by characterizing every component of the network and a stochastic model was designed to describe the hysteretic mammalian gene regulatory network monitored in a lentiviral vector system with single-cell measurements (May et al. 2008). Both models use simple chemical kinetics to depict activation, degradation, inhibition, and formation. Simulation studies by fitting the two models to the experiment data demonstrated that the models had successfully captured the most important characteristic of their corresponding hysteretic mammalian gene regulatory network, whereas the shapes of the simulated loops did not match the experiment data well. Therefore, a more accurate mathematical model is required for modeling the hysteresis in the mammalian gene regulatory networks. Inspired by the similarity in the shape of the experiment hysteresis loop and the one from mathematical hysteresis model, the Bouc–Wen model (Bouc 1976; Wen 1976) is considered to describe the hysteresis in the mammalian gene regulatory networks. The Bouc–Wen model is very popular in mechanical engineering due to the ease of its numerical implementation and its ability to represent a wide range of hysteresis loop shapes. Unfortunately, this original Bouc–Wen model can only generate stable clockwise hysteresis loop while hysteresis is often counterclockwise in biological systems, either the cell cycles or the gene regulatory networks. Therefore, the original Bouc–Wen model was modified in our earlier work to describe the hysteresis in the calcium-mediated ciliary beat frequency dynamics (Qin and Xiang 2011). However, it is not clear whether this model can also characterize the hysteresis in gene regulatory networks. Furthermore, the modified Bouc–Wen model alone cannot produce the transient profile of the step response, which was observed in the experiments (Longo et al. 2010).

It is for the purpose of considering both the steady state hysteretic and transient behaviors of the gene regulatory networks that we propose a more general model based on the modified Bouc–Wen model. By fitting to the experiment data, the hysteresis loops from our model have shown significantly better agreement with the experiment data compared to the KF model and the stochastic model. While the new model shows adequate consistence with the hysteresis in the synthetic mammalian gene regulatory networks, it may not be so convincing that this model can actually describe the hysteresis in the gene networks or it is just good fitting of these specific groups of experiment data in mammalian gene regulatory networks. To investigate the generalizability of the new model, it is applied to reconstruct hysteresis in two other examples of hysteretic gene networks in $E.coli$, which also results in satisfactory agreement with the experiment evidence in the two gene networks with different shapes of hysteresis loops. This suggests that the new model may be a unified framework for modeling the hysteresis in gene regulatory networks, which provides better understanding of the general mechanism that drives hysteresis.

In our earlier work, only simulation results are given whereas the rigorous mathematical proof is missing for the fact that the original Bouc–Wen model cannot generate the counterclockwise hysteresis loop while the modified Bouc–Wen model can generate both types of loops. Furthermore, the mathematical analysis on the dynamical properties of the model is necessary to ensure the bounded-input-bounded-output (BIBO) stability which demonstrates that the model is coherent to the physical property of the real system that it describes. In this paper, we will supply the rigorous mathematical analysis to address these two issues.

The rest of this paper is organized as follows. A general dynamical model including the modified Bouc–Wen hysteresis model as the hysteresis component is developed in Sect. 2. Rigorous mathematical stability analysis on both the original and modified Bouc–Wen hysteresis models is given in Sect. 3, which shows the advantage of the modified Bouc–Wen model. In Sect. 4, simulation results from the following aspects are presented: first, validation of our model on modeling the experimentally observed hysteresis loops at steady state; second, comparison study on the modeling performance with previous models; third, the investigation on the generalizability of the new model for other examples of hysteretic biological networks; fourth, model predictions on the dynamical behavior of the mammalian gene regulatory network. Further discussions are provided in Sect. 5, and Sect. 6 concludes the article.

2 Model Development

2.1 Formulation of Governing Equations

A common feature of the gene regulatory networks with hysteresis is that the inducer activates the hybrid promotor which then drives the gene expression in a positive feedback loop. The mechanism of coupling between the inducer concentration and gene expression is posited in Fig. 3. As shown in Fig. 2, a hysteresis loop at the steady state is observed in the mammalian gene network. The experimental evidence in the literature (Longo et al. 2010) shows that there exists a transient response before the gene expression reaches the steady state. In the following, we propose a general model for systems with hysteresis, which characterizes both the hysteretic and transient behavior of the mammalian gene regulatory network interacted with an inducer.

The concentration of the product of gene expression (output of the gene regulatory network), $y(t)$, is assumed to be governed by the following first order differential equation,

$$T\big(u(t)\big)\frac{dy(t)}{dt} + y(t) = x_2(t) \tag{1}$$

where $x_2(t)$ corresponds to the hysteresis component of the output, which is to be elaborated later. The time constant $T(u(t))$ is a function of the inducer concentration described as

$$T\big(u(t)\big) = \frac{T_c}{1 + \frac{u(t)}{x_r}} \tag{2}$$

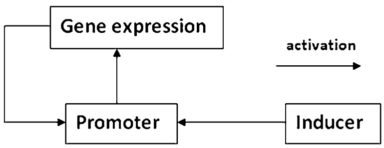


Fig. 3 Configuration of the hysteretic interaction in gene regulatory network

where $u(t)$ is the inducer concentration (input) of the gene regulatory network, T_c is constant and x_r is the threshold value of the inducer concentration for the system to switch from one state to another.

Remark 1 The time constant for first order system is normally a constant. However, in our model, the time constant $T(u(t))$ is designed to be input dependent to be consistent with the experiment evidence (Longo et al. 2010) that the time constant is associated with the input value. For simplicity, we choose the form of $T(u(t))$ expressed in Eq. (2).

The interaction between the inducer concentration $u(t)$ and the promoter activity $x_1(t)$ is described by the following Hill function:

$$x_1(t) = \frac{au(t)^n}{b^n + u(t)^n} \tag{3}$$

where a, b are constant parameters and n is the Hill coefficient describing cooperativity, whose value is system-dependent.

The Hill function is then incorporated into the modified Bouc–Wen model to describe the hysteresis in the gene regulatory networks as follows:

$$\dot{w}(t) = \rho\big(\dot{x}(t) - \sigma\big|\dot{x}(t)\big|w(t) + \lambda x(t)\big|\dot{w}(t)\big| - \gamma w(t)\big|\dot{w}(t)\big|\big) \tag{4}$$

where

$$x(t) = -1 + 2\frac{x_1(t) - c_0}{c_m - c_0}, \tag{5}$$

$$c_0 = \frac{au_{\min}^n}{b^n + u_{\min}^n}, \tag{6}$$

$$c_m = \frac{au_{\max}^n}{b^n + u_{\max}^n}, \tag{7}$$

$$w(t) = -1 + 2\frac{x_2(t) - x_{2\min}}{x_{2\max} - x_{2\min}}. \tag{8}$$

In the above equations, the promotor activity denoted as $x_1(t)$ from Hill function is incorporated into the modified Bouc–Wen model as $x(t)$ is the normalized variable of $x_1(t)$. $w(t)$ is the normalized variable of $x_2(t)$, which denotes the hysteresis component of the system output. $u_{\min}, u_{\max}$ are the minimum and maximum values of the input used in the experiment, and $x_{2\min}, x_{2\max}$ are the minimum and maximum values of the output measured in the experiment, respectively. T_c, x_r, a, b, n, ρ, σ, λ, γ are the model parameters to be determined. All of these parameters are assumed to be positive. The concentration of output signal is mainly regulated by coupled activation and self degradation in both positive and negative feedback loops. On the right-hand side of Eq. (4), the term $\rho(\dot{x}(t) + \lambda|\dot{w}(t)|x(t))$ summarizes nonlinear activation from the hybrid promotor while the term $-\rho(\sigma|\dot{x}(t)|w(t) + \gamma|\dot{w}(t)|w(t))$ summarizes the nonlinear self degradation part (Buchler et al. 2004). These complex interactions are to be elucidated in further experimental study.

Remark 2 Compared to the original normalized Bouc–Wen model in the literature (Ikhouane and Rodellar 2005), the third term on the right side of Eq. (4) is modified, and an additional term, i.e., the fourth term is added (Qin and Xiang 2011). The reason for this modification is that the hysteresis loop generated by the original Bouc–Wen model is clockwise while the hysteresis loop observed in the experiment is clearly counterclockwise. It has been shown by extensive simulation studies that the modified Bouc–Wen model can describe both the clockwise and counterclockwise stable hysteresis loops. In Eq. (4), $\dot{w(t)}$ goes to zero if $\dot{x(t)}$ is zero, that is, $w(t)$ keeps as constant value, which indicates that the modified Bouc–Wen model alone cannot produce the transient time profile of the step response. Therefore, our general model is extended from this modified Bouc–Wen model by adding the transient dynamics in Eq. (1).

Remark 3 Function mapping is more important in the biological model especially for the gene regulatory network as most of the detailed mechanisms of the biochemical processes are unknown but specific and vital functionality can be observed. Hysteresis is such kind of functionality. The key feature of hysteresis in biological network is that the gene expression will take on different values for an increasing of the inducer's concentration than for a decreasing one. This key feature is well characterized by the mathematical terms in Eq. (4), where the absolute values of the derivatives make the form of the equation depend upon the signs of the derivatives, i.e., the increasing and decreasing of the signals.

2.2 Analytical Solution of the Governing Equation for Step Input

In general, for any given input $u(t)$, it is difficult to obtain the analytical solution of general model in Eq. (1). Alternatively, the solution can be obtained numerically using the Runge–Kutta method. In the experiment, the inducer is applied to the gene regulatory network as a step input and the gene expression is measured after 48 hours. For step input, the derivative $\dot{x(t)}$ is delta function. With the special property of delta function, it is possible to obtain the analytical solution of $w(t)$ from Eq. (4). Then the analytical solution of the governing equation for step input can be easily calculated from Eq. (1) with x_2 known from $w(t)$.

The modified Bouc–Wen model in Eq. (4) is complex and nonlinear. In order to derive the solution, we rewrite Eq. (4) as follows:

$$\dot{w(t)} - \rho\lambda x(t)\left|\dot{w(t)}\right| + \rho\gamma w(t)\left|\dot{w(t)}\right| = \rho\left(\dot{x(t)} - \sigma\left|\dot{x(t)}\right|w(t)\right) \tag{9}$$

where

$$\left|\dot{w(t)}\right| = \dot{w(t)}\operatorname{sgn}\left(\dot{w(t)}\right), \qquad \left|\dot{x(t)}\right| = \dot{x(t)}\operatorname{sgn}\left(\dot{x(t)}\right). \tag{10}$$

Substituting Eq. (10) into Eq. (9), we get

$$\dot{w(t)}\left(1 - \rho\lambda x(t)\operatorname{sgn}\left(\dot{w(t)}\right) + \rho\gamma w(t)\operatorname{sgn}\left(\dot{w(t)}\right)\right) = \dot{x(t)}\rho\left(1 - \sigma\operatorname{sgn}\left(\dot{x(t)}\right)w(t)\right) \tag{11}$$

which leads to

$$\dot{w}(t) = \dot{x}(t)\frac{\rho(1-\sigma \operatorname{sgn}(\dot{x}(t))w(t))}{1-\rho\lambda x(t)\operatorname{sgn}(\dot{w}(t))+\rho\gamma w(t)\operatorname{sgn}(\dot{w}(t))}. \tag{12}$$

For the normalized model, proper parameters should be selected during the parameter estimation to avoid the singularity with Eq. (12). In the following, the analytical solution of $w(t)$ when the input is a step function is derived from Eq. (12) and model predicted output $y(t)$ can be obtained from Eq. (1) with $x_2(t)$ known from $w(t)$.

In the process of OFF-to-ON, cells were first cultured without any EM for three days to set the [*SEAP*] in all groups of cells at OFF state then the cells were reseeded under different [*EM*]. Thus, the inducer concentration $u(t)$ is modeled as the following step function:

$$u(t) = \begin{cases} u_{\min}, & t < 0, \\ u_s, & t \geq 0. \end{cases} \tag{13}$$

where $u_{\min}$ is the initial condition that cells were cultured without any inducer in the experiment and u_s is the inducer concentration added into the cell culture medium at the beginning of the 48 hours' experiment.

Both inducer concentration and gene expression level increase from the OFF state and the signs of $\dot{x}(t)$ and $\dot{w}(t)$ are positive. In this case, Eq. (12) can be simplified as follows:

$$\dot{w}(t) = \dot{x}(t)\frac{\rho(1-\sigma w(t))}{1-\rho\lambda x(t)+\rho\gamma w(t)} \tag{14}$$

where

$$x(t) = \begin{cases} -1, & t < 0, \\ x_s, & t \geq 0, \end{cases} \tag{15}$$

where

$$x_s = -1 + 2\frac{c_s - c_0}{c_m - c_0}, \tag{16}$$

$$c_s = \frac{au_s^n}{b^n + u_s^n}, \tag{17}$$

$$c_0 = \frac{au_{\min}^n}{b^n + u_{\min}^n}, \tag{18}$$

$$c_m = \frac{au_{\max}^n}{b^n + u_{\max}^n}. \tag{19}$$

As aforementioned, the experiment input is step signal and its derivative $\dot{x}(t)$ can be expressed as delta function $(x_s + 1)\delta(t)$. To obtain the analytical solution, do the integration on both sides of Eq. (14) and we have

$$\int_0^t \dot{w}(t)\,dt = \int_0^t \dot{x}(t)\frac{\rho(1-\sigma w(t))}{1-\rho\lambda x(t)+\rho\gamma w(t)}\,dt, \tag{20}$$

$$w(t) - w(0) = \int_0^t (x_s + 1)\delta(t) \frac{\rho(1 - \sigma w(t))}{1 - \rho\lambda x(t) + \rho\gamma w(t)} dt \tag{21}$$

where $w(0) = -1$ and $t > 0$.

Here, we have to deal with the integration of $\delta(x)f(x)$ where $f(x)$ is a step function jumping at the origin and we cannot directly use the selecting property of the delta function as $f(x)$ is discontinuous at the origin. Before we go on deriving the solution, we consider a more general case where $f(x)$ is any function that is discontinuous at the origin and continuous elsewhere. Assume the left limit and the right limit of the function $f(x)$ at the origin are $f(0^-)$ and $f(0^+)$. For simplicity, we assume that the delta function is represented by the limit of the square pulse function as follows:

$$\delta(x) = \begin{cases} \lim_{n\to\infty} n, & -\frac{1}{2n} \le x \le \frac{1}{2n}, \\ 0, & \text{for all other } x. \end{cases} \tag{22}$$

It follows that

$$\int_{-\infty}^{+\infty} \delta(x)f(x)\,dx = \int_{-\frac{1}{2n}}^{\frac{1}{2n}} \delta(x)f(x)\,dx = \int_{-\frac{1}{2n}}^{0} \delta(x)f(x)\,dx + \int_{0}^{\frac{1}{2n}} \delta(x)f(x)\,dx. \tag{23}$$

Using the definition of delta function in Eq. (22), we have

$$\int_{-\infty}^{+\infty} \delta(x)f(x)\,dx = \lim_{n\to\infty} \int_{-\frac{1}{2n}}^{0} nf(x)\,dx + \lim_{n\to\infty} \int_{0}^{\frac{1}{2n}} nf(x)\,dx. \tag{24}$$

Applying the first mean value theorem for integration, we have

$$\int_{-\infty}^{+\infty} \delta(x)f(x)\,dx = \frac{f(0^-) + f(0^+)}{2}. \tag{25}$$

Applying the above formula to Eq. (21), we get

$$w(t) = w(0) + (x_s + 1)\frac{\frac{\rho(1-\sigma w(0))}{1-\rho\lambda(-1)+\rho\gamma w(0)} + \frac{\rho(1-\sigma w(t))}{1-\rho\lambda x_s+\rho\gamma w(t)}}{2}. \tag{26}$$

In the process of ON-to-OFF, cells were first cultured with maximum inducer concentration used in the experiment and then reseeded under different inducer concentration. Thus the inducer concentration $u(t)$ is step input as following:

$$u(t) = \begin{cases} u_{\max}, & t < 0, \\ u_s, & t \ge 0, \end{cases} \tag{27}$$

where $u_{\max}$ is the initial condition that cells were cultured with maximum inducer concentration in the experiment and u_s is the inducer concentration under which the cells were reseeded for the 48 hours' experiment.

Both the inducer concentration and gene expression level decrease from the ON state and the signs of $\dot{x(t)}$ and $\dot{w(t)}$ are negative. In this case, Eq. (12) can be simplified as follows:

$$\dot{w(t)} = \dot{x(t)}\frac{\rho(1+\sigma w(t))}{1+\rho\lambda x(t) - \rho\gamma w(t)} \tag{28}$$

where

$$x(t) = \begin{cases} 1, & t < 0, \\ x_s, & t \geq 0, \end{cases} \tag{29}$$

where x_s can be obtained similarly from Eq. (16) to Eq. (19).

Similarly, do the integration on both sides of Eq. (28) and we have

$$\int_0^t \dot{w(t)}\,dt = \int_0^t \dot{x(t)}\frac{\rho(1+\sigma w(t))}{1+\rho\lambda x(t) - \rho\gamma w(t)}\,dt, \tag{30}$$

$$w(t) - w(0) = \int_0^t (x_s - 1)\delta(t)\frac{\rho(1+\sigma w(t))}{1+\rho\lambda x(t) - \rho\gamma w(t)}\,dt, \tag{31}$$

$$w(t) = w(0) + (x_s - 1)\frac{\frac{\rho(1+\sigma w(0))}{1+\rho\lambda - \rho\gamma w(0)} + \frac{\rho(1+\sigma w(t))}{1+\rho\lambda x_s - \rho\gamma w(t)}}{2} \tag{32}$$

where $w(0) = 1$ and $t > 0$.

Both Eq. (26) and Eq. (32) can be formulated into the quadratic equation $Aw^2 + Bw + C = 0$ and each of them has two roots $w_1 = \frac{-B-\sqrt{B^2-4AC}}{2A}$ and $w_2 = \frac{-B+\sqrt{B^2-4AC}}{2A}$. In the model, $w(t)$ is normalized into $[-1, 1]$. According to the simulation result, only one root is located within the required range for $w(t)$. More specifically, from OFF to ON, the root w_2 is within the required range while w_1 is far away from the range, thus w_2 is chosen as the final analytical solution of the steady state output for the process from OFF to ON. Similarly, w_1 is chosen as the final analytical solution of the steady state output for the process from ON to OFF.

Once $w(t)$ is obtained from the modified Bouc–Wen model, $x_2(t)$ can be calculated from Eq. (8). Since $x_2(t)$ is a step function and the time constant $T(u(t))$ is also a constant, the analytical solution of the governing equation for both OFF-to-ON and ON-to-OFF can be easily derived from Eq. (1) and Eq. (2) as follows:

$$y(t) = x_2(t) + \big(y(0) - x_2(t)\big)e^{-\frac{t}{T}}, \quad t > 0 \tag{33}$$

where

$$T = \frac{T_c}{1 + \frac{u_s}{x_r}}. \tag{34}$$

2.3 Collection of the Experimental Data

The experimentally measured steady state value of the gene expression is used to estimate the parameter values of our model for further numerical studies. To be more specific, two sets of experiment data denoted as Cell A and Cell B were collected from Kramer and Fussenegger's experiment. The experiment data for Cell A and Cell B are extracted from Fig. 5A and Fig. 6A in the literature (Kramer and Fussenegger 2005), respectively.

2.4 Parameter Identification

In the general model, there are two parameters T_c and x_r, which can be estimated directly from the experiment. The experiment data were collected after 48 hours and it takes about five times of T_c to approach the steady state thus the value of T_c is estimated to be $\frac{48}{5}$. x_r is the threshold value of the inducer concentration for the gene expression switching from OFF-to-ON and the value observed in the experiment is 1000 ng/mL [*EM*]. Now only the seven parameters for the hysteresis component, a, b, n, ρ, σ, δ, γ are to be identified. These parameter values can be obtained using the analytical solution of Eq. (1), i.e., Eq. (33).

As widely used in information technology, the basic idea of parameter identification is using optimization algorithm to find the optimal parameter values which minimize the difference between the model-predicted output and the experiment value defined by

$$E = \sum_{m=1}^{M} \left(y_{\text{pred}}(m) - y(m)\right)^2 \tag{35}$$

where M is the total number of experiment samples, $y(m)$ denotes the observed steady state value of mth experiment gene expression measured after sufficient time, $y_{\text{pred}}(m)$ denotes the corresponding mth predicted steady state value from the general model expressed by Eq. (1) to Eq. (8). With any given set of parameters, $y_{\text{pred}}(m)$ for the step input can be calculated from the analytical solutions of the equations. We adopted the Nelder–Mead simplex algorithm (Nelder and Mead 1965) to solve the optimization problem in Eq. (35) iteratively. Matlab optimization toolbox *fmincon* is applied to implement the aforementioned parameter identification strategy. The estimated values of the parameters for the general model are shown in Table 1.

3 The Stability Analysis

In practice, mathematical modeling is often implemented by the black-box approach that given a set of experimental input-output data, one tries to find the optimal model parameters such that the output of the model matches the experimental data. However, it may happen that the mathematical model presents a good match with the experimental real data for a specific input, but does not necessarily keep significant physical properties which are inherent to the real data, independently of the exciting input (Ikhouane and Rodellar 2005).

It can be readily seen from the general equation that the sufficient and necessary condition for the boundedness of the output $y(t)$ in Eq. (1) is that $x_2(t)$ is bounded. Therefore, in the following, we only present mathematical stability analysis of the hysteresis component and both the original and the modified Bouc–Wen model are

Table 1 Parameter values of the general model for hysteretic mammalian gene regulatory network in Cell A and Cell B

Parameter	Description	Unit	Cell A	Cell B
T_c	Constant in the general model	hours	9.6	9.6
x_r	Threshold value of inducer concentration	ng/mL	1000	1000
$u_{\min}$	Minimum value of inducer concentration	ng/mL	0	0
$u_{\max}$	Maximum value of inducer concentration	ng/mL	1000	1500
$x_{2\min}$	Minimum value of gene expression	mU/L	0	0
$x_{2\max}$	Maximum value of gene expression	mU/L	350	64
a	Constant in Hill function		162.2	236
b	Constant in Hill function	ng/mL	441.6	431.5
n	Hill coefficient		2	5
ρ	Parameter in modified Bouc–Wen model		0.5	0.2
σ	Parameter in modified Bouc–Wen model		0.8	0.01
λ	Parameter in modified Bouc–Wen model		3	5.5
γ	Parameter in modified Bouc–Wen model		1.2	1.02

analyzed on their bounded-input-bounded-output (BIBO) stability property for comparison. First, we find the parameter regions such that the model is BIBO stable for both models via Lyapunov stability analysis; then, in the BIBO stable parameter regions, we investigate the ability of both models to generate clockwise and counterclockwise loops by analyzing the convexity of the loop or the sign of the second-order derivative.

3.1 The Bounded-Input-Bounded-Output (BIBO) Stability Analysis

The BIBO stability for the original Bouc–Wen model is analyzed by Lyapunov function in (Ikhouane and Rodellar 2005). Now we are going to do a similar analysis on the modified Bouc–Wen model in Eq. (4).

We choose the quadratic Lyapunov function,

$$V(t) = \frac{w(t)^2}{2}, \tag{36}$$

and we have

$$\dot{V}(t) = \dot{w}(t)w(t). \tag{37}$$

Substituting Eq. (4) into above equation yields

$$\dot{V}(t) = \rho\big(\dot{x}(t) - \sigma\big|\dot{x}(t)\big|w(t) + \lambda\big|\dot{w}(t)\big|x(t) - \gamma\big|\dot{w}(t)\big|w(t)\big)w(t). \tag{38}$$

It follows that

$$\dot{V}(t) \le \rho\big(\big|\dot{x}(t)\big|\big|w(t)\big| - \sigma\big|\dot{x}(t)\big|w(t)^2 + \lambda\big|\dot{w}(t)\big|\big|x(t)\big|\big|w(t)\big| - \gamma\big|\dot{w}(t)\big|w(t)^2\big), \tag{39}$$

$$\dot{V}(t) \le \rho\big(\big|\dot{x}(t)\big|\big(\big|w(t)\big| - \sigma w(t)^2\big) + \big(\lambda\big|x(t)\big|\big|w(t)\big| - \gamma w(t)^2\big)\big|\dot{w}(t)\big|\big). \tag{40}$$

Thus, $\dot{V}(t) \le 0$ if the following two conditions are satisfied,

$$\big|w(t)\big| - \sigma w(t)^2 \le 0, \tag{41}$$

and

$$\lambda\big|x(t)\big|\big|w(t)\big| - \gamma w(t)^2 \le 0. \tag{42}$$

The above two conditions imply that

$$\big|w(t)\big| \ge \max\left(\frac{1}{\sigma}, \frac{\lambda|x(t)|}{\gamma}\right), \tag{43}$$

which indicates that whenever $|w(t)|$ is bigger than $\max(\frac{1}{\sigma}, \frac{\lambda|x(t)|}{\gamma})$, we have $\dot{V}(t)$ is non-positive. Hence, as long as the input $x(t)$ is bounded, $w(t)$ is also bounded. Therefore, the modified Bouc–Wen model is BIBO stable when all the parameters are positive.

Now we are going to investigate that inside the BIBO stable region, whether the hysteresis loops from both the original and modified Bouc–Wen models are clockwise or counterclockwise. Here, we assume $\frac{d^2w(t)}{dx(t)^2}$ exists and $\dot{x}(t) \neq 0$.

3.2 The Original Bouc–Wen Model Can Only Generate a Clockwise Loop

The normalized Bouc–Wen model (Ikhouane and Rodellar 2005) is described as follows:

$$\dot{w}(t) = \rho\big(\dot{x(t)} - \sigma\big|\dot{x}(t)\big|\big|w(t)\big|^{n-1}w(t) + (\sigma - 1)\dot{x}(t)\big|w(t)\big|^n\big). \tag{44}$$

Consider the following four subcases based on the signs of $w(t)$ and $\dot{x}(t)$,

$$w(t) \ge 0, \qquad \dot{x}(t) > 0, \qquad \frac{d^2w(t)}{dx(t)^2} = \rho^2\big(1 - w(t)^n\big)\big(-nw(t)^{n-1}\big); \tag{45}$$

$$\begin{aligned} &w(t) \le 0, \qquad \dot{x}(t) > 0, \\ &\frac{d^2w(t)}{dx(t)^2} = \rho^2\big(1 + (2\sigma - 1)\big(-w(t)\big)^n\big)n(2\sigma - 1)\big(-\big(-w(t)\big)^{n-1}\big); \end{aligned} \tag{46}$$

$$\begin{aligned} &w(t) \ge 0, \qquad \dot{x}(t) < 0, \\ &\frac{d^2w(t)}{dx(t)^2} = \rho^2\big(1 + (2\sigma - 1)w(t)^n\big)n(2\sigma - 1)w(t)^{n-1}; \end{aligned} \tag{47}$$

$$w(t) \le 0, \qquad \dot{x}(t) < 0, \qquad \frac{d^2w(t)}{dx(t)^2} = \rho^2\big(1 - \big(-w(t)\big)^n\big)n\big(-w(t)\big)^{n-1}. \tag{48}$$

It stems from the stability analysis that the upper bound is $|w(t)| \le 1$ (Ikhouane and Rodellar 2005). Thus, it can be concluded that for $\sigma \ge 0.5$, we have

$$\dot{x}(t) > 0, \qquad \frac{d^2w(t)}{dx(t)^2} \le 0, \tag{49}$$

and

$$\dot{x}(t) < 0, \qquad \frac{d^2 w(t)}{dx(t)^2} \geq 0. \tag{50}$$

This means that when the input increases, the curve is concave and when the input decreases, the curve is convex, which indicates that the hysteresis loop is clockwise.

For $0 < \sigma < 0.5$, we have $-1 < 2\sigma - 1 < 0$ and

$$w(t) \geq 0, \qquad \dot{x}(t) > 0, \qquad \frac{dw(t)}{dx(t)} = \rho\big(1 - w(t)^n\big); \tag{51}$$

$$w(t) \leq 0, \qquad \dot{x}(t) > 0, \qquad \frac{dw(t)}{dx(t)} = \rho\big(1 + (2\sigma - 1)\big(-w(t)\big)^n\big); \tag{52}$$

$$w(t) \geq 0, \qquad \dot{x}(t) < 0, \qquad \frac{dw(t)}{dx(t)} = \rho\big(1 + (2\sigma - 1)\big(w(t)\big)^n\big); \tag{53}$$

$$w(t) \leq 0, \qquad \dot{x}(t) < 0, \qquad \frac{dw(t)}{dx(t)} = \rho\big(1 - \big(-w(t)\big)^n\big). \tag{54}$$

It is also obvious that when $w(t) \geq 0$,

$$\rho\big(1 - w(t)^n\big) < \rho\big(1 + (2\sigma - 1)w(t)^n\big); \tag{55}$$

and when $w(t) \leq 0$,

$$\rho\big(1 - \big(-w(t)\big)^n\big) < \rho\big(1 + (2\sigma - 1)\big(-w(t)\big)^n\big). \tag{56}$$

This indicates that when $w(t) \leq 0$, the value of $\frac{dw(t)}{dx(t)}$, i.e., the slope of the curve $w(x)$, is bigger for $\dot{x}(t) > 0$ than that for $\dot{x}(t) < 0$, which means that if both start from the lowest point, the curve $w(x)$ increases faster for $\dot{x}(t) > 0$ than that for $\dot{x}(t) < 0$. In other words, the curve $w(x)$ for $\dot{x}(t) > 0$ is above that for $\dot{x}(t) < 0$ for those points near the lowest point. How about the points further away from the minimum?

It is noted from Eqs. (51) to (54) that the slopes of the curve $w(x)$ only depend upon the value of w, and for any $w_1(t) < w_2(t)$, we have

$$\rho\big(1 - \big(-w_1(t)\big)^n\big) < \rho\big(1 + (2\sigma - 1)\big(-w_2(t)\big)^n\big). \tag{57}$$

It implies that when $w(t) \leq 0$ for any point on the curve $w(x)$ for $\dot{x}(t) > 0$, its slope is always bigger than that for $\dot{x}(t) < 0$ as long as its value is larger. It follows immediately that when $w(t) \leq 0$ if both start from the lowest point, the curve $w(x)$ for $\dot{x}(t) > 0$ is above that for $\dot{x}(t) < 0$ since its slope is always bigger at every point.

In a similar fashion, we can also show that when $w(t) \geq 0$ if both start from the highest point, the curve $w(x)$ for $\dot{x}(t) > 0$ is above that for $\dot{x}(t) < 0$ since its slope is always smaller and, therefore, its value decreases slower than that for $\dot{x}(t) < 0$.

Overall, we have demonstrated that the curve $w(x)$ for $\dot{x}(t) > 0$ is above that for $\dot{x}(t) < 0$ if they share the same maximal and minimal points which indicates that the hysteresis loop is always clockwise.

Therefore, the hysteresis loops from the Bouc–Wen model can only be clockwise and the original Bouc–Wen model can not generate the counterclockwise hysteresis loops observed in biological systems such as the gene networks.

3.3 Clockwise and Counterclockwise Hysteresis Loops from the Modified Bouc–Wen Model

To simplify the discussion, we will only consider the situation that the sign of $w\dot(t)$ is always the same as that of $x\dot(t)$. In other words, $x(t)$ and $w(t)$ will increase and decrease together, which is true for most of the real systems. Under this condition, the modified Bouc–Wen model in Eq. (4) can be rewritten as follows:

$$\frac{dw(t)}{dx(t)} = \begin{cases} \frac{\rho(1-\sigma w(t))}{1-\rho\lambda x(t)+\rho\gamma w(t)}, & \dot{x}(t) > 0, \\ \frac{\rho(1+\sigma w(t))}{1+\rho\lambda x(t)-\rho\gamma w(t)}, & \dot{x}(t) < 0. \end{cases} \tag{58}$$

Then we have

$$\frac{d^2w(t)}{dx(t)^2} = \begin{cases} \frac{\rho^2(1-\sigma w(t))(-\rho\gamma-\sigma+\lambda+\rho\lambda(\sigma-\lambda)x(t)+\rho\lambda\gamma w(t))}{(1-\rho\lambda x(t)+\rho\gamma w(t))^3}, & \dot{x}(t) > 0, \\ \frac{\rho^2(1+\sigma w(t))(\rho\gamma+\sigma-\lambda+\rho\lambda(\sigma-\lambda)x(t)+\rho\lambda\gamma w(t))}{(1+\rho\lambda x(t)-\rho\gamma w(t))^3}, & \dot{x}(t) < 0. \end{cases} \tag{59}$$

Since $x\dot(t)$ and $w\dot(t)$ have same sign, we can get

$$\frac{dw(t)}{dx(t)} > 0. \tag{60}$$

Combining with Eq. (58), we have

$$\frac{\rho^2(1-\sigma w(t))}{(1-\rho\lambda x(t)+\rho\gamma w(t))^3} > 0, \tag{61}$$

and

$$\frac{\rho^2(1+\sigma w(t))}{(1+\rho\lambda x(t)-\rho\gamma w(t))^3} > 0. \tag{62}$$

Therefore, the sign of the second-order derivative depends mainly on the signs of the following two terms:

$$-\rho\gamma - \sigma + \lambda + \rho\lambda(\sigma - \lambda)x(t) + \rho\lambda\gamma w(t), \tag{63}$$

and

$$\rho\gamma + \sigma - \lambda + \rho\lambda(\sigma - \lambda)x(t) + \rho\lambda\gamma w(t). \tag{64}$$

It has been proved that any positive parameter values will make the model BIBO stable, thus it is very easy to choose the proper parameters to make the above terms either negative or positive. In other words, the modified Bouc–Wen model can generate both clockwise and counterclockwise loops which has also been verified by extensive simulation studies.

4 Simulation Results

4.1 The Step Response of the Mammalian Gene Regulatory Network

In Kramer and Fussenegger's experiment, two different clones CHO-HYST42 (Cell A) and CHO-HYST44 (Cell B) were chosen. The two sets of experiment data from Cell A and Cell B were used to estimate the model parameter for Cell A and Cell B, respectively, and simulation studies on the model validation and prediction are presented after model identification. The step response of the gene regulatory network in Cell B is simulated from our model.

As shown in Fig. 4, in the process of OFF to ON, cells were first cultured without any EM for three days to set the [*SEAP*] in all groups of cells at OFF state then the cells were reseeded under different [*EM*] and the [*SEAP*] in all of them will increase from the OFF state with addition of different magnitude of [*EM*]. To be more specific, the steady state value of [*SEAP*] is higher with bigger [*EM*] signal comparing between different groups from OFF-to-ON. Similarly, as shown in Fig. 5, in the process of ON to OFF, cells were first cultured with sufficient EM for three days to set the [*SEAP*] in all groups of cells at ON state then the cells were reseeded with different [*EM*] and the [*SEAP*] in all of them will decrease from the ON state with addition of different magnitude of [*EM*]. To be more specific, the [*SEAP*] is lower with smaller [*EM*] signal comparing between different groups from ON-to-OFF. The steady state values of [*SEAP*] measured after 48 hours are marked with the circles in the two figures. It is evident that the simulation results tally well with the experiment data.

4.2 Hysteresis Loops at Steady State

4.2.1 Model Validation on Mammalian Hysteretic Gene Regulatory Networks

Figure 6 shows hysteretic EM-SEAP kinetics in the mammalian gene regulatory network in Cell A. The circles and the black dots connected by the solid line corre-

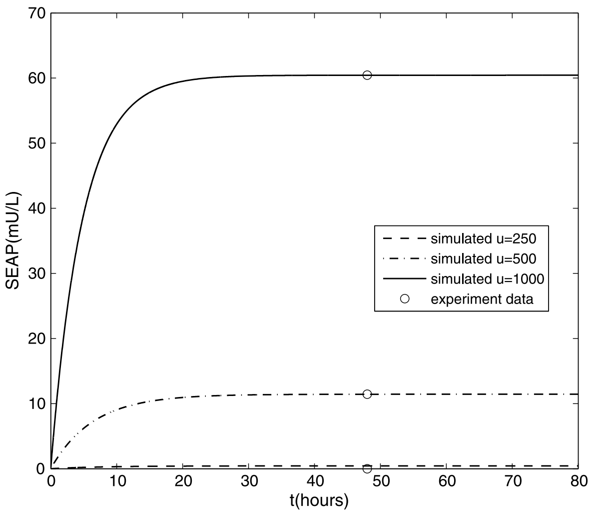


Fig. 4 The transient time profile of the process from OFF to ON in Cell B with EM concentration at 250, 500, 1000 ng/mL

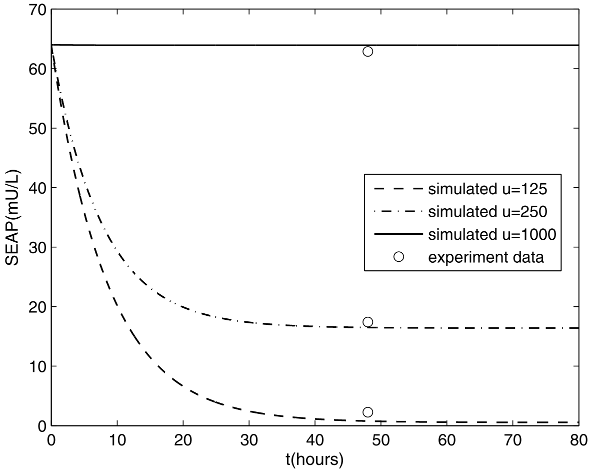


Fig. 5 The transient time profile of the process from ON to OFF in Cell B with EM concentration at 125, 250, 1000 ng/mL

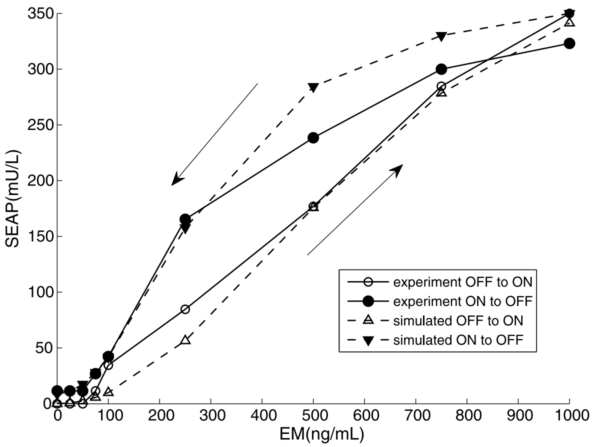


Fig. 6 Model validation test on EM-SEAP kinetics in Cell A

spond to the experimental gene expression switching from OFF-to-ON and ON-to-OFF, respectively, based on Kramer and Fussenegger's experiment result (Kramer and Fussenegger 2005). The up and down triangle are the simulated gene expression switching from OFF-to-ON and ON-to-OFF, respectively, fitted by our model. The arrows that go up and down denote the processes of OFF-to-ON and ON-to-OFF.

It is obvious from Fig. 6 that the hysteresis loop from our model is counterclockwise which demonstrates that our model has captured the most important feature of the experiment result, that is, higher [*EM*] is required for gene expression switching from OFF-to-ON than that from ON-to-OFF. The simulated hysteresis loop shows adequate consistence with the experiment data. As exemplified by this study, our general model has captured the cooperativity relationship between [*EM*] and [*SEAP*] correctly, which verifies the capability and accuracy of the mathematical model to emulate natural gene expression behavior with hysteresis.

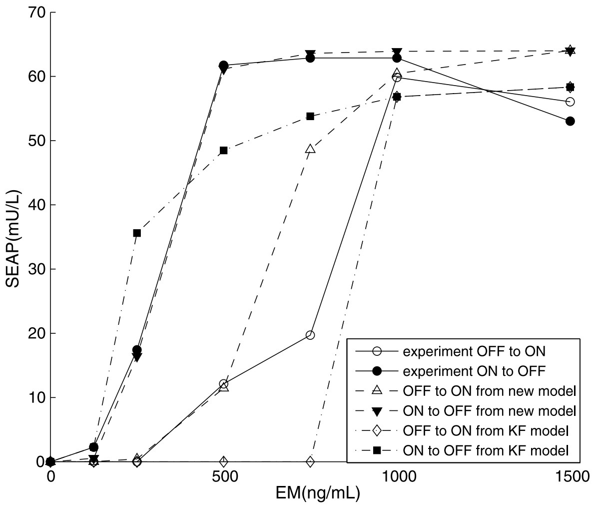


Fig. 7 Comparison with previous model on EM-SEAP kinetics in Cell B

4.2.2 Comparison with Previous Models

Figure 7 shows the comparison study on the hysteretic EM-SEAP kinetics in the mammalian gene regulatory network in Cell B. Similarly, the circle and the black dots connected by the solid line correspond to the experiment gene expression switching from OFF-to-ON and ON-to-OFF, respectively, based on the experiment result (Kramer and Fussenegger 2005). The arrows that go up and down denote the processes of OFF-to-ON and ON-to-OFF. The dashed line is the hysteresis loop fitted by the new model of the EM-SEAP kinetics. The dashdot line is the simulated hysteresis loop from Kramer and Fussenegger's model (KF model) (Kramer and Fussenegger 2005). It can be readily seen from Fig. 7 that while both the hysteresis loops from the two models have captured the most important feature of the hysteretic gene regulatory network as they are both counterclockwise, the fitted result from our model shows significantly better agreement with the experiment result than KF model, which indicates that our model is more accurate at least from phenomenological point of view.

Further comparison study on another mammalian gene network is described in Fig. 10 with details in Appendix 1 as well as the model generalizablity in two *E.coli* gene networks in Fig. 11 and Fig. 12 in Appendix 2.

4.3 Model Predictions on the Dynamical Behavior of Mammalian Gene Network

4.3.1 Dynamical Response to Step Input Beyond the Switching Threshold

Induced by $[EM]$ above the threshold value, the gene expression in all cells will eventually reach a steady state at high level. In other words, there is switch-like activation in the network that the gene expression at OFF state in all cells are to be switched to ON state. To find out whether the inducer concentration would affect the activation time of the gene network prior to the ON state, we are motivated to investigate the difference between the transient time profiles of the gene expression induced by

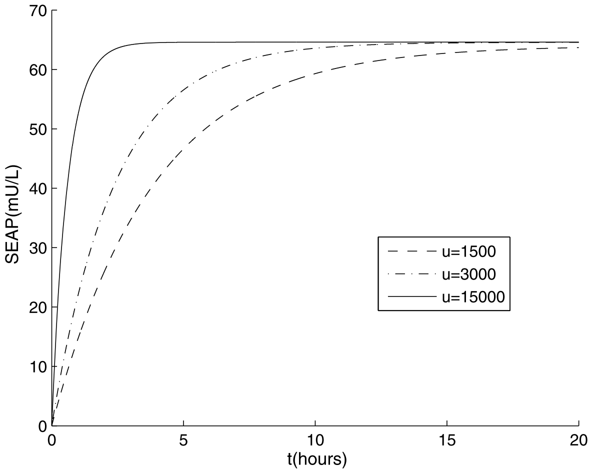


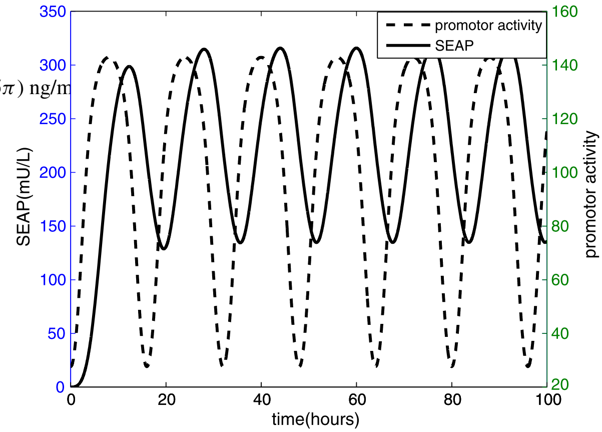
Fig. 8 Predicted time profile of [*SEAP*] in Cell B induced with 1500, 3000, 15 000 ng/mL EM

different [*EM*] above the threshold value. The threshold value of [*EM*] for [*SEAP*] to switch from OFF-to-ON is 1000 ng/mL as observed from the experiment result in Fig. 2. It can be readily seen from Fig. 8 that beyond the threshold of [*EM*] that all the gene expression would be at ON state in the steady state, the activation time of [*SEAP*] induced with 15 000 ng/mL [*EM*] is much shorter than that induced with 3000 ng/mL [*EM*] and the one induced with 1500 ng/mL [*EM*] has the longest activation time to reach the steady state. It is evident from this simulation study that the activation time prior to the ON state is smaller with larger inducer concentration beyond the threshold value. Unfortunately, there is no direct experimental evidence for us to validate the simulation prediction on this specific mammalian gene regulatory network. However, there is experiment observation in another mammalian gene regulatory network which shows that activation behavior becomes more coherent and it is faster to reach the steady state value as the strength of the positive feedback loop becomes stronger (Longo et al. 2010). In our mammalian gene regulatory network, the antibiotic EM induces the positive feedback loop between SEAP and hybrid promotor and the strength of the positive feedback loop becomes stronger as [*EM*] increases. The simulation prediction is consistent with this indirect experiment result as the time for [*SEAP*] to reach the steady state value becomes shorter with larger [*EM*].

4.3.2 Dynamical Response to Periodic Input

Figure 9 describes the oscillation behavior of [*SEAP*] in the gene regulatory network predicted from the general model with the onset of pulsatile antibiotic EM concentration, $u(t) = 700 + 500\sin(0.125\pi t + 1.5\pi)$ ng/mL. With known parameter values in Table 1, though it is difficult to obtain the analytical solution, the model predicted [*SEAP*] can be calculated numerically from Eq. (1) to Eq. (8) using the well-known fourth- order Runge–Kutta method (Stoer and Bulirsch 1993). During the initial period of the onset of the pulsatile input signal [*EM*], the [*SEAP*] goes up in a step response manner and later the dynamical behavior of [*SEAP*] shows oscillation. The hybrid promotor activity which is represented by $x_1(t)$ is also depicted in the figures. It is evident from Fig. 9 that there is obvious time lag between output signal

Fig. 9 Predicted oscillation behavior of [*SEAP*] in Cell A induced with EM $u(t) = 700 + 500\sin(0.125\pi t + 1.5\pi)$ ng/ml



[*SEAP*] and the hybrid promotor activity of the gene regulatory network. This is consistent with the experiment observations of the oscillations in the positive feedback loop in the mammalian genetic oscillator (Tigges et al. 2009). Although a number of experiment observations have validated the genetic oscillations from *E.coli* to mammalian and human synthetic gene regulatory network (Elowitz and Leibler 2000; Portle et al. 2009; Tigges et al. 2009; Toettcher et al. 2010), the dynamical prediction from this simulation study remains to be verified by direct experiment on this specific synthetic mammalian gene regulatory network.

5 Discussion

A number of investigations by in vitro experiments on the synthetic gene regulatory networks have established that hysteresis phenomenon occurs in the interaction between the output signal (gene expression) and the input signal (inducer) of the synthetic mammalian gene regulatory network (Longo et al. 2010; Kramer and Fussenegger 2005; May et al. 2008). It is for the purpose of considering both the steady state hysteresis phenomenon and also the transient dynamical behavior in the mammalian gene regulatory network that a general model is proposed by combination of both the hysteresis component and the transient dynamics.

In Kramer and Fussenegger's experiment, the mammalian gene regulatory network consists of a transgene and transactivator (TA) cotranscribed by TA's cognate promoter, repressed by constitutive expression of a macrolide-dependent transcriptional silencer, whose activity is modulated by the macrolide antibiotic EM. The threshold value of [*EM*] to switch [*SEAP*] from OFF to ON is higher than that from ON to OFF. In this situation, the threshold is not constant but differs according to which state the gene regulatory network is moving from and depends on the historical EM concentration (Greber and Fussenegger 2007). The two computational models, KF model proposed by Kramer and Fussenegger (2005) and the stochastic model (May et al. 2008), were designed to describe the hysteresis component in synthetic mammalian gene regulatory networks. Both models had successfully captured the essential feature of the hysteretic gene regulatory network, that is, bigger inducer is required to switch the gene expression from OFF-to-ON than that from ON-to-OFF. However, the simulated hysteresis loops from the two models did not fit the experiment data very accurately. In this paper, our general model is developed by combining the modified Bouc–Wen model as the hysteresis component and adding the transient dynamics to generate the transient response of step input. Simulation results from our model (Fig. 6 and Fig. 7) exhibit adequate agreement with the experimental results in Cell A and Cell B. This verifies that our general model is valid and accurate for modeling the hysteresis in the mammalian gene regulatory network. Comparison study shows that this model fits the experiment data significantly better than the previous ones (the dashed dot line in Fig. 7 and Fig. 10 in Appendix 1). The hysteresis loop from the new model is more accurate to describe the hysteresis in the mammalian gene regulatory network at least from phenomenological point of view.

It is interesting to note that the shape of the simulated hysteresis loop from our model is quite different for Cell A from that for Cell B. This is consistent with the experiment observation that the gene switch in Cell A shows graded behavior while that in Cell B is bistable. From this point of view, it actually shows the advantage of our model that it can model a wide variety of different hysteresis loops which makes good sense as the profiles of gene regulatory networks are usually different from each other. To further investigate the generalizability of the model on describing hysteresis in different biological networks, the new model is applied to describe the hysteresis in two other examples of hysteretic gene regulatory networks in *E.coli* (see Appendix 2). Despite the fact that the shapes of the hysteresis loops in these examples are strikingly different from each other, simulation results from our model exhibit adequate consistence with the experimental results of both hysteresis gene regulatory networks. Thus, our model has the potential to provide a unified framework for the various hysteretic biological networks as it can describe a wide range of different hysteresis responses accurately.

While the mathematical model can provide good data fitting performance, it is also important for the model to guarantee dynamical property that is coherent to real systems that the model describes. In practice, the response of the real system to an limited input cannot blow up, which indicates that the bounded-input-bounded-output stability is fundamental for real system. Correspondingly, the mathematical model should also be ensured of the BIBO stability. It may happen that the model provides a good match for some specific input but it can give unstable output for other input signal. A Lyapunov analysis has been carried out to verify that the modified Bouc–Wen model is BIBO stable no matter what the input signal is. Furthermore, the rigorous mathematical proof has been given to show that the original Bouc–Wen model can only generate the clockwise hysteresis loops and it cannot describe the counter clockwise hysteresis loops observed frequently in biological systems. This demonstrates the advantage of the modified Bouc–Wen model that it can generate both clockwise and counterclockwise hysteresis loops and it can also produce a wide variety of hysteresis loop shapes.

Modeling the hysteresis of five different gene regulatory networks in this paper may provide better understanding of the general mechanism that drives hysteresis. The ubiquity of the positive feedback in all the five hysteretic gene regulatory networks suggests that it is the core topology of the network to achieve the functionality of hysteresis. In systems science, feedback refers to the output of the system being regulated by the output itself. In Eq. (4) of our model, the output of the equation depends on itself coupled with other components on the right-hand side of the equation. The output concentration is regulated by complex interactions between both input and output itself's concentration and rate, which is to be verified by further experimental study. Briefly speaking, $\rho(\dot{x}(t) + \lambda|\dot{w}(t)|x(t))$ can be regarded as the nonlinear activation part while $-\rho(\sigma|\dot{x}(t)|w(t) + \gamma|\dot{w}(t)|w(t))$ can be considered as the nonlinear self degradation part (Buchler et al. 2004). Therefore, the new model is a mathematical description that couples with both positive feedback and also the negative feedback. These feedback loops interacting with each other results in the universal ability of the new model to generate large varieties of hysteresis loops in the gene networks. In other words, the model captures the key element of the system structure responsible for the hysteresis in gene regulatory networks. With a focus on the particular biological functionality, hysteresis, the new model is much simpler than those involved dozens of biochemical reactions inside the biological networks as it only considers the essential interactions between two variables, the input and the output. In this sense, the biological systems are highly complex while simple mathematical models may provide the system level quantitative understanding of the complex biological systems.

The simulation predictions from the general model also provide some insights on the dynamical behavior of the gene regulatory networks. The first investigation is that above the threshold where the gene expression will all eventually reach the ON state, what the difference is in the time profile of the gene expression prior to the steady state when the gene network is induced with different magnitude of the inducer concentration and the second consideration is how the gene regulatory network responds to the pulsatile input. The former one is addressed by the simulation prediction result in Fig. 8 that the time for [*SEAP*] to reach the steady state value decreases with the increasing of the inducer level above the threshold as the strength of the positive feedback loop becomes stronger with larger [*EM*], which is consistent with the indirect experiment evidence (Longo et al. 2010). It can be readily seen from Fig. 9 that the gene regulatory network oscillates in a pulsatile responsive manner with obvious time delay. The oscillation behavior speculated from the simulation study based on the general model is also consistent with the experiment observations in the gene oscillator (Elowitz and Leibler 2000; Portle et al. 2009; Tigges et al. 2009; Toettcher et al. 2010). However, these predictions remain to be verified by direct experiment observations in this particular mammalian gene regulatory network.

6 Conclusion

The main contribution of this paper is proposing a general model for hysteresis observed in gene regulatory networks by incorporating the modified Bouc–Wen model as the hysteresis component and also the transient dynamics. Rigorous mathematical analysis is presented to demonstrate the good dynamical property of the bounded-input-bounded-output stability which is coherent with the essential physical property of real systems and also shows the advantage of the modified Bouc–Wen model in the aspect of describing both clockwise and counterclockwise hysteresis loops while the original one can only generate clockwise hysteresis loops. Simulation studies have evidently shown the hysteresis loops from our model are consistent with the experiment observations in three mammalian gene regulatory networks and two $E.coli$ gene regulatory networks. This demonstrates the ability and accuracy of the mathematical model to emulate natural gene expression behavior with hysteresis. Comparison study has illustrated that this model fits the experiment data significantly better than previous ones in the literature. The successful modeling of hysteresis in all the five hysteretic gene regulatory networks demonstrates the general capability of the mathematical model on describing different hysteresis phenomena which suggests that the new model may be a unified framework for modeling the hysteresis in biological networks and provide better understanding of the general mechanism that drives the hysteretic function.

Acknowledgements The research reported here was supported by NUS Academic Research Fund R-263-000-483-112, the National Natural Science Foundation of China (Grant No. 11172060) and the Fundamental Research Funds for the Central Universities in China.

Appendix 1: Model Validation on Another Mammalian Gene Network

Another synthetic mammalian positive feedback network was monitored in a lentiviral vector system with single-cell measurements (May et al. 2008). The enhanced green fluorescent protein (eGFP) mutant is regulated by the inducer doxycycline. The eGFP positive cell population is designated with HIGH and the eGFP negative cell population is designated with LOW. The experimental observed hysteresis loop between the percentage of the high cells and the concentration of the doxycycline is shown as the solid line in Fig. 10. A stochastic Markov process was presented based on the simplified reaction scheme as a mathematical description of the synthetic hysteretic gene regulatory circuit (May et al. 2008). The simulation result from the stochastic model shown as the dash-dot line in Fig. 10 predicted the existence of hysteresis which was verified by later experiment. However, the simulated hysteresis loop from the stochastic model did not match the experiment data very well. In this paper, we investigate whether our model can describe the hysteresis in this synthetic mammalian gene regulatory network more accurately. The concentration of doxycycline is the input ($u(t)$) and the percentage of the high cells is the output ($y(t)$). Due to the missing information on the time constant and threshold value in the experiment, the output value cannot be directly obtained from the general model. However, the experimental hysteresis loop is measured at steady state. When the time goes to infinity, the output value in Eq. (1) is constant for step input and its value is equal to the hysteresis component. Therefore, it is reasonable for us to obtain the steady state output value by deriving the value of the hysteresis component using the modified Bouc–Wen model in Eq. (3) to Eq. (8). The experiment data is extracted from Fig. 5B (May et al. 2008) and the whole parameter identification procedure is the same as that for Cell A and Cell B. The estimated parameter values are listed in Table 2 and the simulated loop from our model is shown as a dashed line in Fig. 10. Comparison results from the stochastic model and the new model shown in Fig. 10 demonstrate

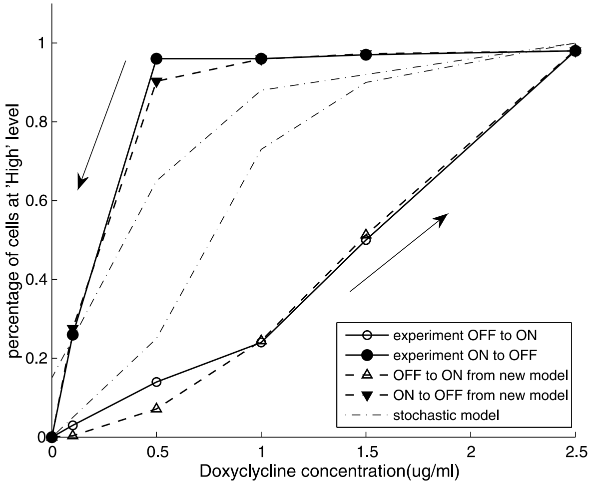


Fig. 10 Hysteretic interactions between Dox and eGFP in synthetic mammalian gene network

Table 2 Parameter values of the modified Bouc–Wen model for the hysteretic gene networks in Fig. 10, Fig. 11, Fig. 12

Parameter	Description	Fig. 10	Fig. 11	Fig. 12
$u_{\min}$	Minimum input value	0	0	0
$u_{\max}$	Maximum input value	2.5 μg/ml	4	4
$x_{2\min}$	Minimum output value	0	0	0
$x_{2\max}$	Maximum output value	0.98	1	2800 *Miller Units*
a	Constant in Hill function	1.2	1.75	2818.2
b	Constant in Hill function	0.534 μg/ml	2.01	1.5
n	Hill coefficient	2	3	5
ρ	Parameter in modified Bouc–Wen model	0.19	0.69	0.7
σ	Parameter in modified Bouc–Wen model	0.01	0.001	0.01
λ	Parameter in modified Bouc–Wen model	5	1	0.8
γ	Parameter in modified Bouc–Wen model	0.285	0.184	0.01

that our model shows substantially better agreement with the experiment result compared to the stochastic model though both models have captured the hysteretic feature in the gene regulatory network. Thus, it is evident to see the ability and accuracy of the new model to describe the hysteretic response in the mammalian gene regulatory network.

Appendix 2: Model Generalizability on *E.coli* Hysteretic Gene Regulatory Networks

In this section, two more examples of hysteretic gene networks in *E.coli* are utilized to test the generalizability of our model for modeling the hysteresis in gene regulatory network. Similarly, the steady state output value is obtained by deriving the hysteresis

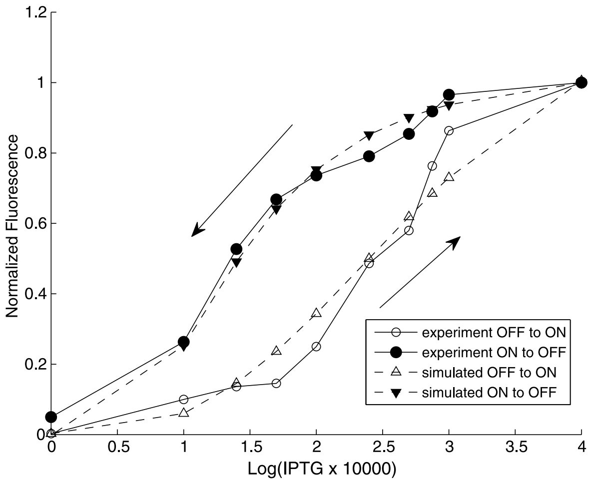


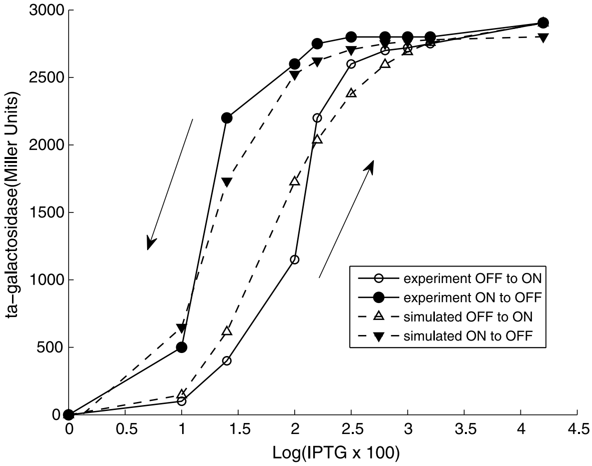
Fig. 11 Hysteretic interactions between IPTG and GFP in synthetic gene regulatory network of $E.coli$

component value due to the missing information on the time constant and threshold value. The two sets of experiment data are extracted from Fig. 3A (Maeda and Sano 2006) and Fig. 3B (Chang et al. 2010) in the literature. The input values in both gene network are normalized as their order of magnitude vary sharply. The whole parameter identification procedure is the same as that for Cell A and Cell B. The estimated parameter values are in Table 2.

The chemical inducer IPTG is widely used as it cannot be metabolized by $E.coli$. A chromosomally expressing lactose repressor protein LacI, inhibits the expression of the gene coding for green fluorescence protein (GFP) by binding to the lactose operator. In the synthetic $E.coli$ gene regulatory network, IPTG inactivates LacI to regulate $[GFP]$ in a positive feedback loop. The interconnections between experiment measured $[GFP]$ and $[IPTG]$ shows hysteresis (Maeda and Sano 2006). By taking the normalized inducer concentration $\mathrm{Log}([IPTG] \times 10000)$ as the input $(u(t))$ and $[GFP]$ as the output $(y(t))$, the new model is employed to characterize the hysteresis in the $E.coli$ synthetic gene regulatory network. The simulated hysteresis loop shown in Fig. 11 demonstrates adequate agreement with the experiment data. This verifies the ability of the mathematical model to describe the hysteretic interaction between GFP and IPTG in the $E.coli$ gene network.

IPTG induces the transcription of the gene coding for beta-galactosidase. The beta-galactosidase provides positive feedback that drives its own expression induced by IPTG, which inactivates repressor (Chang et al. 2010). By taking the normalized inducer concentration $\mathrm{Log}([IPTG] \times 100)$ as the input $(u(t))$ and the concentration of beta-galactoside as the output $(y(t))$, our general model is used to describe the hysteresis in $E.coli$ synthetic gene regulatory network. Figure 12 shows the simulated hysteresis loop between beta-galactoside and the IPTG in this synthetic gene regulatory network. The strong consistence between the simulated result and the experiment data again demonstrates that the model can also describe the hysteresis in the synthetic gene regulatory network of $E.coli$.

Fig. 12 Hysteretic interactions between IPTG and beta-galactosidase in synthetic gene regulatory network of *E.coli*



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